What Offline Observations Can Establish About Adaptive Feedback in Machine Markets

Machine Markets project
Authorship and contribution review pending

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Abstract

We report a reproducible next-stage validation of a machine-markets research release. The study separates measurement uncertainty, adaptive-policy comparisons, and operational loss. In an independently specified synthetic market, a small neural policy was trained by deterministic population search and compared on paired held-out tapes with a fixed no-trade rule and a hand-written adaptive non-neural controller. Across three training seeds and 128 held-out episodes per seed, the mean neural-minus-adaptive gain was -0.022 account units, far below the predeclared 0.5-unit effect threshold; no AI-specific advantage is supported. A separate observation benchmark shows that a constrained noisy/dependent condition had 0.790 interval coverage, 0.075 wrong-confident calls, and 0.905 unresolved calls. In a difficult synthetic risk regime, a 50% execution throttle reduced the nonterminal event rate from 0.090 to 0.000 while reducing mean drawdown to 0.010. These are controlled synthetic findings, not real-market probabilities or safety guarantees. Public Binance OHLCV and historical depth-band/trade-tape data are used for a chronological descriptive transfer layer; queue-level replay, external replication, and independent publication review remain open.

1 Scope and research question

The central question is deliberately narrower than the original ambition: what can observations and declared architectural information establish about adaptive feedback at a stated economic cost, and what remains necessary to assess operational loss? We distinguish three targets that are often conflated: local signal stability, finite-observation inference, and financially induced loss or halt events. A numerical test of one target is not evidence for the others.

This paper is a new evidence layer on top of the frozen Machine Markets release. The four historical study packages and their checksums are preserved. New code is under `studies/next_stage_validation/`; its outputs are not pooled with the original synthetic denominators.

2 Information contract and environment

The new environment is a separate CPU-only implementation with explicit price, cash, inventory, spread, depth, impact, partial fill, fee, margin, drawdown, and halt logic. At each step an agent observes a seven-component public vector: price return, normalized inventory, equity change, spread, depth, margin ratio, and normalized time. Shock values and future prices are not exposed. The action is a requested signed quantity in $[-1, 1]$.

The primary confirmation configuration uses a 48-step horizon, 100 units of initial cash, maximum order 0.5, maximum inventory 2, spread 0.02, depth 5, impact 0.04, volatility 0.01, fee rate 0.001, and partial-fill parameter 0.15. The environment is a controlled synthetic mechanism, not a claim to reproduce an exchange.

Table 1: Paired matched-agent results. Values are mean account gains.

Seed	No trade	Adaptive	Neural	Neural - adaptive
601	0.000	-0.093	-0.183	-0.090
602	0.000	-0.093	-0.116	-0.023
603	0.000	-0.093	-0.045	0.048

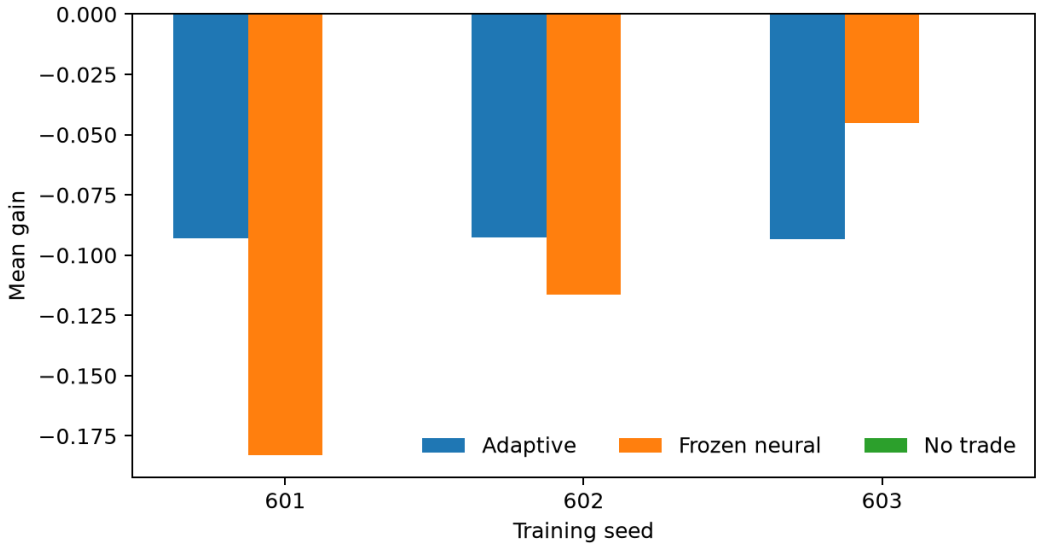


Figure 1: Held-out paired gains by training seed. The no-trade rule has no transaction loss in this environment; that is a baseline, not a profitable strategy claim.

3 Matched policies and training

The fixed policy submits no order. The adaptive non-neural policy is a declared linear controller of current return and inventory. The trained policy is a seven-input, eight-hidden-unit tanh network. Its weights are optimized by a deterministic cross-entropy population search on development tapes, then frozen for held-out evaluation. Training uses eight iterations, population 24, three development episodes per candidate, and seeds 601–603. Each training seed is reported separately; its 128 held-out episodes are paired across all three policies using the same tape.

The primary effect is the paired neural-minus-adaptive gain. A 0.5-unit mean improvement on a 100-unit account was declared the minimum meaningful effect before the confirmation run. Event rate, final equity, and maximum drawdown are secondary outcomes. This design can test a narrow policy-family comparison; it cannot identify a universal AI effect or a causal effect in real markets.

4 Measurement study

The observation benchmark simulates a latent episode process, adds measurement noise to regressors and responses, optionally introduces within-episode AR dependence, and masks observations. The slope estimator includes an intercept. Uncertainty is estimated by resampling complete episodes. Rows from a long trace are never treated as independent units. Coverage is simulation-calibrated for the declared law, not an exact theorem.

The exact independent condition behaves as expected under this simulation. Once measurement noise, dependence, and a smaller budget are combined, the constrained condition has 0.790 coverage and 0.075 wrong-confident labels, while 0.905 of replicates remain unresolved at the declared decision threshold. The result is a limitation of the implemented procedure, not evidence for a universal impossibility theorem.

Table 2: Observation diagnostics. Coverage, decisiveness, wrong-confidence, and unresolved fractions are separate metrics.

Condition	Coverage	Decisive	Wrong confident	Unresolved
exact_independent	1.000	1.000	0.000	0.000
noisy_independent	0.835	1.000	0.000	0.000
noisy_dependent	0.795	1.000	0.000	0.000
noisy_dependent_missing	0.805	1.000	0.000	0.000
constrained_noisy_dependent	0.790	0.095	0.075	0.905

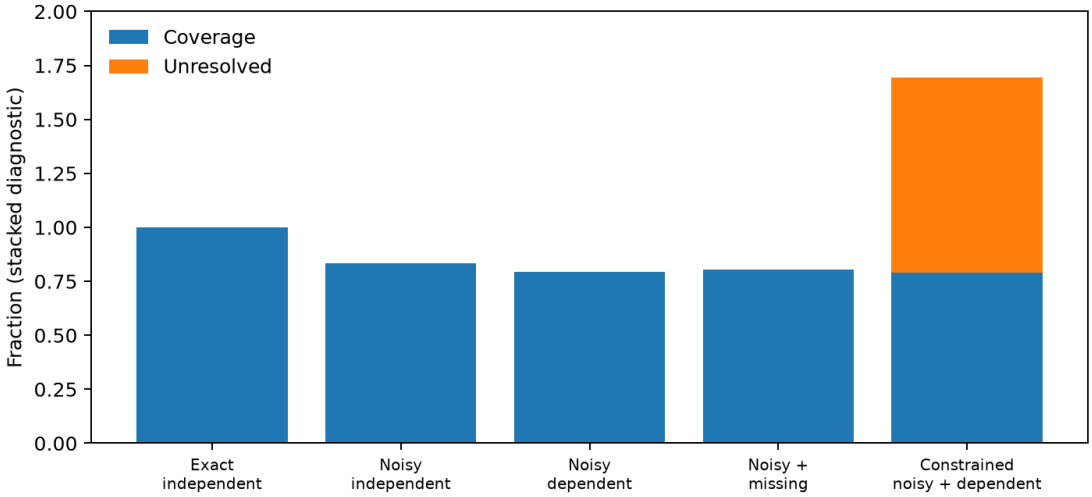


Figure 2: Observation coverage and unresolved diagnostics. The stacked display is descriptive; the two fractions are not additive evidence of a probability model.

5 Operational-risk and intervention study

The operational event is any nonterminal outcome: halt, default, margin breach, or drawdown halt. A normal terminal settlement is not an event. The difficult regime uses 50 units of initial cash, maximum order and inventory 1, volatility 0.02, impact 0.30, fee rate 0.002, margin rate 0.40, maintenance margin 0.10, and drawdown limit 0.05. The three interventions were fixed before running 256 paired held-out tapes: no intervention, a 50% order throttle, and inventory- triggered liquidation.

The throttle lowers the local event rate and mean drawdown in this declared single-account environment. Because there is no external dealer or second account, the experiment cannot test loss shifting. Therefore it supports only a conditional synthetic result, not an operational safety recommendation.

6 Public-data transfer boundary

We load three public Binance monthly kline archives: BTCUSDT one-minute, ETHUSDT one-hour, and SOLUSDT one-hour data from January 2025. The split is chronological 60% training, 20% validation, and 20% test. The resulting descriptive counts are shown below.

We additionally downloaded and checksum-verified 93 daily Binance “bookDepth“ archives and 93 matched “aggTrades“ archives for BTCUSDT, ETHUSDT, and SOLUSDT over January 2025. The chronological held-out summaries are shown below.

These files provide aggregate depth bands and executions, not historical level-2 queue state. They do not contain cancellations, queue priority, spread, or submitted-order outcomes, and the signed impact quantity is descriptive rather than causal. The corpus therefore completes a historical depth-band and trade-tape calibration layer but does not establish a calibrated queue-level exchange replay environment. Full L2/fill

Table 3: Synthetic operational interventions.

Policy	Event rate	Gain	Drawdown
none	0.090	-0.939	0.023
throttle	0.000	-0.387	0.010
liquidate	0.043	-0.838	0.021

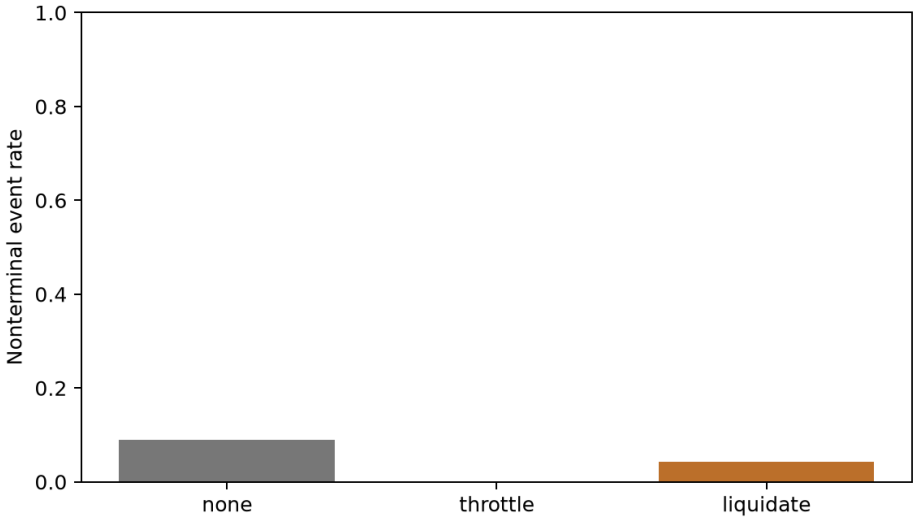


Figure 3: Nonterminal event rates under the three fixed interventions.

calibration remains externally blocked until authorized event history or an equivalent validated interface is available.

7 Relation to the frozen release and failure cases

The original release's internal tests and algebra checks remain the baseline; they do not become external replication by being rerun. The next-stage package preserves the following adverse findings: stronger full-horizon baselines can outperform summary procedures, high-noise errors remain, omitted slow states and false rate floors can produce false-stable labels, endpoint diversity is not uniformly beneficial, and binding constraints can cause halts despite local signal stability. The new results add a trained-policy null result and a constrained observation failure mode.

8 What is established

Within the declared synthetic environment and seeds, the trained neural policy does not clear the predeclared effect threshold against the adaptive non-neural baseline. The evidence therefore supports a general adaptive-market and measurement-limitation framing, not an AI-specific claim. The episode-block benchmark demonstrates how noisy/dependent observations can lower coverage and increase unresolved or wrong-confident decisions under a finite budget. The intervention experiment demonstrates one conditional synthetic mechanism by which execution throttling reduces local event rates.

9 Limitations and open gates

There is no independent external replication report, no authenticated deployment/runtime attestation, no queue-level/fill calibration, no external operational validation, and no independent novelty, authorship, ethics,

Table 4: Public OHLCV pilot split.

Asset	Train	Validation	Test	Return volatility
BTCUSDT-1m-2025-01	26784	8928	8928	0.00080
ETHUSDT-1h-2025-01	446	148	150	0.00734
SOLUSDT-1h-2025-01	446	148	150	0.01238

Table 5: Historical public depth-band and aggregate-trade calibration targets.

Symbol	Test days	Snapshots/day	Trades	Flow ACF	Return vol.
BTCUSDT	7	2880	10725827	0.521	0.00003
ETHUSDT	7	2880	11111798	0.561	0.00003
SOLUSDT	7	2880	3443026	0.257	0.00006

or publication review. The trained policy is small and synthetic; its failure to show an advantage is not evidence that all AI systems lack one. No live trading, market perturbation, paid data, or unauthorized access was used. The paper is internally reproducible at the declared level but is not presented as publication-complete or safety-validated.

10 Data and code availability

The frozen release remains under `paper/` and `studies/`. Run the new local confirmation package with:

```
python3.11 -m pytest -q studies/next_stage_validation/tests
python3.11 studies/next_stage_validation/scripts/run_all.py \
  --output studies/next_stage_validation/results --episodes 128
python3.11 studies/next_stage_validation/scripts/run_interventions.py \
  --output studies/next_stage_validation/results --episodes 256
python3.11 studies/next_stage_validation/scripts/run_observation_benchmark.py \
  --output studies/next_stage_validation/results --replicates 200
python3.11 studies/next_stage_validation/scripts/analyze_results.py
python3.11 studies/next_stage_validation/scripts/render_paper_artifacts.py
python3.11 studies/next_stage_validation/scripts/run_l2_calibration.py \\
  --depth-root PATH_TO_BOOKDEPTH_ROOT \\
  --trades-root PATH_TO_AGGTRADES_ROOT \\
  --output studies/next_stage_validation/results
```

The OHLCV acquisition manifest is in the repository's data directory; the historical depth and trade archive manifest and its exact paths are documented in `docs/L2_CALIBRATION_REPORT.md`. The raw L2 corpus is retained on T9 and is not copied into GitHub. Generated results are versioned separately from the original master checksum manifest.

Contribution and review status

The contribution record, AI-use disclosure, permissions, independent review, and venue-specific rights are not finalized in this artifact. This section is intentionally explicit so that a generated PDF is not mistaken for a completed publication process.